

**Question 1 (15 Marks)****Marks**

(a) Sketch the curve  $(x + 1)(y - 2) = 1$

**2**

(b) Write the function in part (a) in the form  $y = f(x)$ . Hence, or otherwise, sketch the curve

(i)  $y = f(x - 2)$

**1**

(ii)  $y = \sqrt{f(x)}$

**2**

(c) Evaluate

(i)  $\int_2^3 \frac{4}{(x-1)(4-x)} dx$

**3**

(ii)  $\int_0^1 \frac{dx}{9-4x^2}$

**2**

(d) (i) The point  $P$  moves so that the sum of its distances from the points  $(-2, 0)$  and  $(2, 0)$  is 6 units. Prove that the equation of the locus of  $P$  is  $\frac{x^2}{9} + \frac{y^2}{5} = 1$ .

**4**

(ii) Find the eccentricity of this locus.

**1****Question 2 (15 Marks) START A NEW PAGE****Marks**

(a) The hyperbola  $H$  is given by the equation  $\frac{x^2}{16} - \frac{y^2}{25} = 1$ .

(i) Write down the equations of the asymptotes.

**2**

(ii)  $P$  is an arbitrary point with coordinates  $(4\sec\theta, 5\tan\theta)$ . Show that  $P$  lies on  $H$ .

**2**

(iii) Prove that the tangent to  $H$  at  $P$  has equation  $\frac{x\sec\theta}{4} - \frac{y\tan\theta}{5} = 1$ .

**3**

(iv) This tangent cuts the asymptotes in  $L$  and  $M$ . Prove that  $LP = PM$ .

**3**

**Question 2 Continued****Marks**

(b) Let  $n$  be a positive integer and  $I_n = \int_1^2 (\ln x)^n dx$ .

(i) Prove that  $I_n = 2(\ln 2)^n - nI_{n-1}$ .

**2**

(ii) Hence, evaluate  $\int_1^2 (\ln x)^3 dx$ .

**3****Question 3 (15 Marks) START A NEW PAGE****Marks**

(a) Find  $\int \frac{x+1}{x^2+1} dx$ .

**2**

(b) On the attached sheet, you are given the curve of  $y = f(x)$ .  
Sketch neatly on separate diagrams

(i)  $y = |f(x)|$

**1**

(ii)  $y = \frac{1}{f(x)}$

**2**

(c) (i) Evaluate  $\int_{-\pi}^{\pi} x \cos x dx$ .

**2**

(ii) Evaluate  $\int_0^{\frac{\pi}{2}} \cos^5 \theta d\theta$ .

**3**

(d)  $F$  is the point  $(4, 0)$  and  $d$  is the line  $x = 1$ .  $M$  is the foot of the perpendicular from a variable point  $P$  to  $d$ , and  $P$  moves so that  $FP = 2PM$ .

(i) Derive the equation of the locus of  $P$ .

**3**

(ii) Find the acute angle between the asymptotes to the nearest degree.

**2**

**Question 4 (15 Marks)    START A NEW PAGE****Marks**

(a) Use the substitution  $t = \tan \frac{x}{2}$  to prove that  $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \sin x} = \frac{\pi}{3\sqrt{3}}$ .

**4**

(b) (i) Show that  $\int_0^{2a} f(x) dx = \int_0^a \{f(x) + f(2a - x)\} dx$ .

**3**

(ii) Hence, or otherwise evaluate  $\int_0^{\pi} \frac{x dx}{2 + \sin x}$ .

**3**

(c) (i) Evaluate  $\int_0^{\frac{\pi}{4}} \sec \theta d\theta$ .

**2**

(ii) Hence, show that  $\int_0^{\frac{\pi}{4}} \sec^3 \theta d\theta = \frac{1}{2} [\sqrt{2} + \ln(\sqrt{2} + 1)]$

**3**

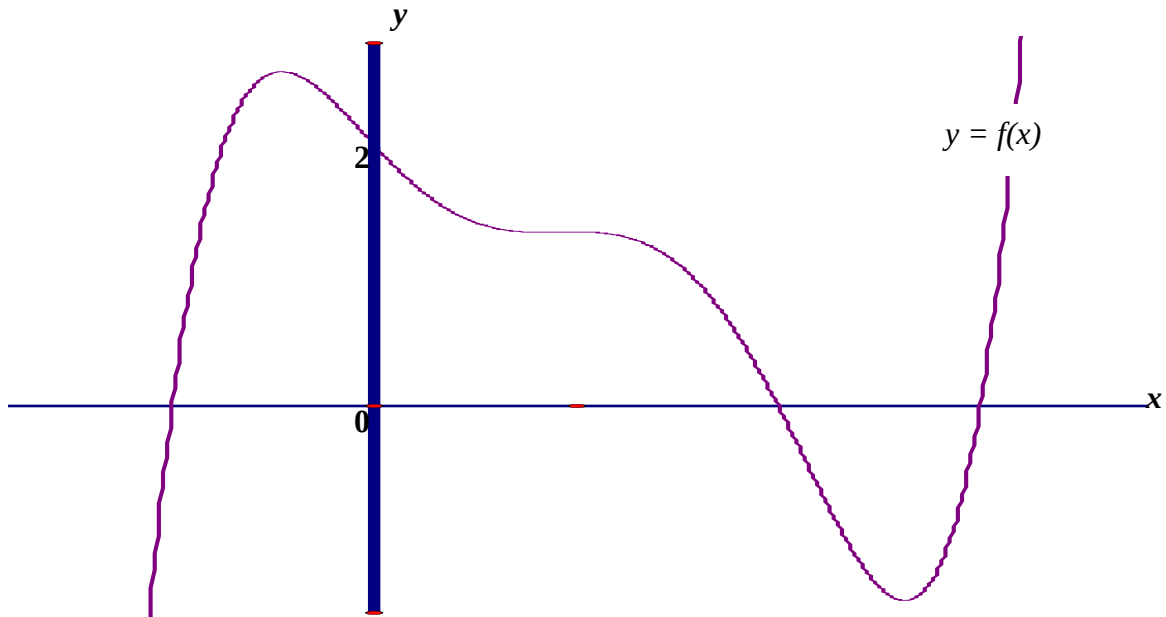
**~ END OF ASSESSMENT~**

**Question 3 Part (b)**  
**HAND IN WITH YOUR ANSWERS**

Sketch neatly on separate diagrams

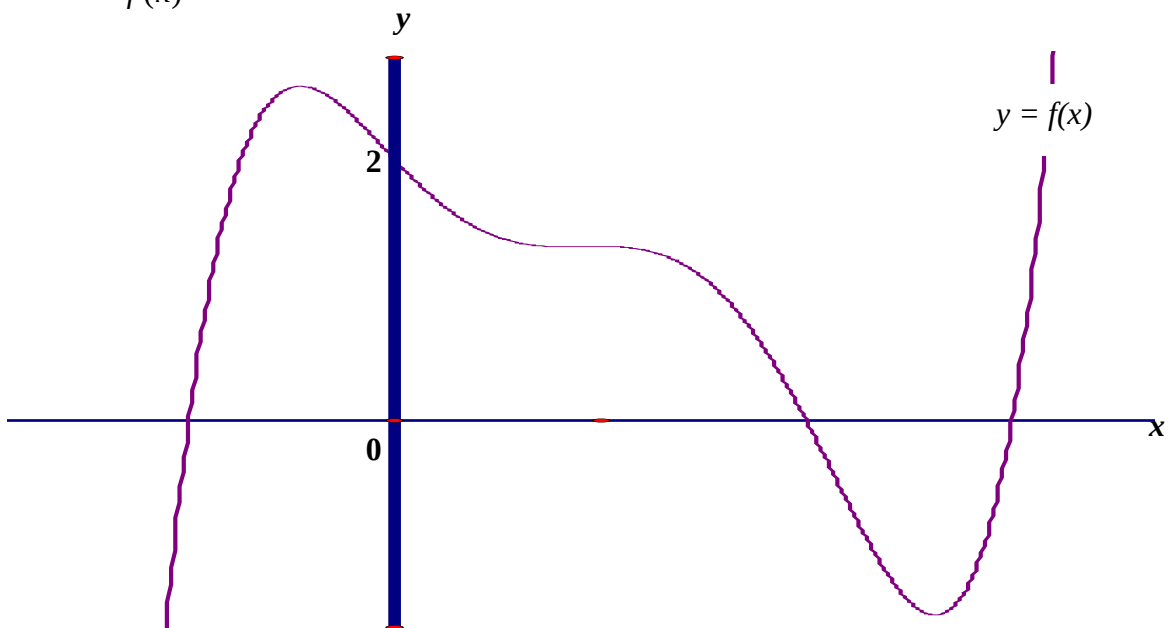
(i)  $y = |f(x)|$

1



(ii)  $y = \frac{1}{f(x)}$

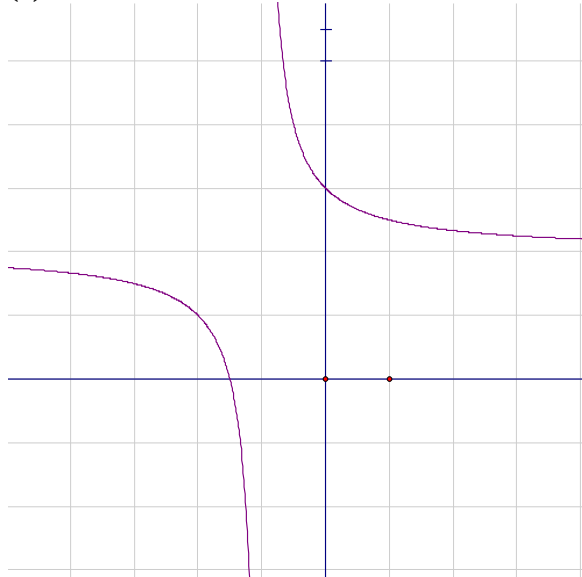
2



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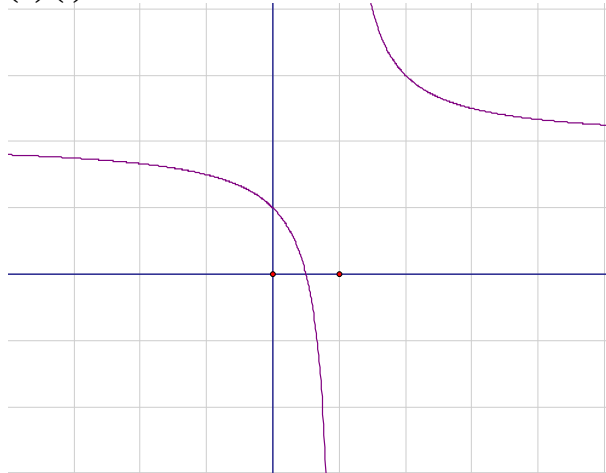
**Question 1**

(a)



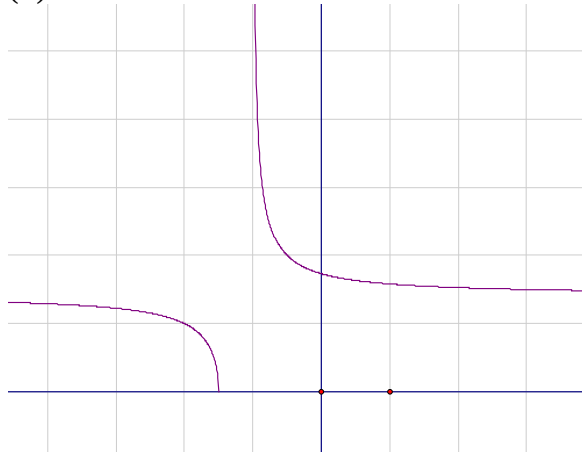
Asymptotes  $x = -1$  &  $y = 2 \rightarrow$  1 mark  
Shape  $\rightarrow$  1 Mark

(b) (i)



$y = f(x - 2)$  is a horizontal shift of  $y = f(x)$  2 units to the right.  $\rightarrow$  1 Mark  
(Asymptotes at  $y = 2$  &  $x = 1$ )

(ii)



$$y = \sqrt{f(x)}$$

Asymptote at  $x = -1$  &  $y = \sqrt{2}$ ,  $\rightarrow$  ½ Mark

$y$  - intercept =  $\sqrt{3}$  &

$x$  - intercept =  $-1.5 \rightarrow$  1 Mark

No graph below  $x$  - axis  $\rightarrow$  ½ Mark

Shape is relatively similar to the original function  $\therefore$  no marks awarded

Q1 (c)

$$(i) \int_2^3 \frac{4}{(x-1)(4-x)} dx$$

$$= \frac{4}{3} \int_2^3 \left( \frac{1}{x-1} + \frac{1}{4-x} \right) dx$$

By Partial Fractions

$$= \frac{4}{3} [\ln(x-1) - \ln(4-x)]_2^3$$

$$= \frac{4}{3} [(\ln 2 - \ln 1) - (\ln 1 - \ln 2)]$$

$$= \frac{4}{3} [2 \ln 2] = \frac{8 \ln 2}{3}$$

Correct values of A and B → 1 Mark

Correct integration → 1 Mark

Correct answer → 1 Mark

$$(ii) \int_0^1 \frac{dx}{9-4x^2}$$

$$= \frac{1}{6} \int_0^1 \left( \frac{1}{3+2x} + \frac{1}{3-2x} \right) dx$$

$$= \frac{1}{6} \left[ \frac{\ln(3+2x)}{2} - \frac{\ln(3-2x)}{2} \right]_0^1$$

$$= \frac{1}{12} [(\ln 5 - \ln 3) - (\ln 1 - \ln 3)]$$

$$= \frac{\ln 5}{12}$$

**This is not a inverse Trigonometry Question!**  
No marks awarded if used incorrect method.

Correct use of partial fractions & integration → 1 Mark

Correct evaluation → 1 Mark

(d) (i) P is (x, y) then by data

$$\sqrt{(x+2)^2 + y^2} + \sqrt{(x-2)^2 + y^2} = 6$$

$$\text{i.e. } 12\sqrt{x^2 + 4x + 4 + y^2} = 36 - 8x$$

$$\therefore 9(x^2 - 4x + 4 + y^2) = 81 - 36x + 4x^2$$

$$\therefore 5x^2 + 9y^2 = 45 \Rightarrow \frac{x^2}{9} + \frac{y^2}{5} = 1$$

→ 1 Mark

→ 1 Mark

→ 1 Mark

→ 1 Mark

(ii) Using  $b^2 = a^2(1 - e^2)$  since locus is an ellipse where  $a = 3$  and  $b = \sqrt{5}$ , we get

$$e = \frac{2}{3}.$$

→ 1 Mark

**Question 2**

(a) (i) Eqn. of asymptotes  $\frac{x^2}{16} - \frac{y^2}{25} = 0$

$$\therefore y = \pm \frac{5x}{4}$$

(ii) If  $P$  is  $(4\sec\theta, 5\tan\theta)$  then

$$\begin{aligned} \text{LHS} &= \frac{x^2}{16} - \frac{y^2}{25} \\ &= \frac{16\sec^2\theta}{16} - \frac{25\tan^2\theta}{25} \\ &= \tan^2\theta + 1 - \tan^2\theta \\ &= 1 = \text{RHS} \end{aligned}$$

$\therefore P$  lies on hyperbola  $H$ .

(iii)  $\frac{dy}{dx} = \frac{5\sec\theta}{4\tan\theta}$  at  $P$ .

$\therefore$  eqn. of tangent at  $P$  is

$$y - 5\tan\theta = \frac{5\sec\theta}{4\tan\theta}(x - 4\sec\theta)$$

$$4\tan\theta y - 5x\sec\theta = 20(\tan^2\theta - \sec^2\theta)$$

$$\therefore \frac{x\sec\theta}{4} - \frac{y\tan\theta}{5} = 1$$

(iv) tangent cuts  $y = \frac{5x}{4}$  when

$$\frac{x\sec\theta}{4} - \frac{5x}{4} \cdot \frac{\tan\theta}{5} = 1$$

$$\therefore x = \frac{4}{\sec\theta - \tan\theta} \text{ and } y = \frac{5}{\sec\theta - \tan\theta}.$$

$$\therefore L \left[ \frac{4}{\sec\theta - \tan\theta}, \frac{5}{\sec\theta - \tan\theta} \right]$$

Tangent cuts  $y = -\frac{5x}{4}$  at

$$M \left[ \frac{4}{\sec\theta + \tan\theta}, \frac{-5}{\sec\theta + \tan\theta} \right]$$

If  $P$  is the midpoint of  $LM$  then  $LP = PM$ .

→ 1 Mark

→ 1 Mark

→ 1 Mark

Showing LHS = RHS → 1 Mark

Correct differential → 1 Mark

Correct eqn. of tangent → 1 Mark

Correct rearrangement to get required answer → 1 Mark

Correct coordinates of  $L$  → ½ Mark

Coordinates of  $M$  → ½ Mark

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                             |
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| <p> <math>\therefore x</math>- coordinate of midpoint of <math>LM</math><br/> <math>= \frac{1}{2} \left[ \frac{4}{\sec \theta - \tan \theta} + \frac{4}{\sec \theta + \tan \theta} \right]</math><br/> <math>= \frac{1}{2} \left[ \frac{8 \sec \theta}{\sec^2 \theta - \tan^2 \theta} \right]</math><br/> <math>= 4 \sec \theta.</math> </p> <p> <math>y</math>- coordinate of midpoint of <math>LM</math><br/> <math>= \frac{1}{2} \left[ \frac{5}{\sec \theta - \tan \theta} + \frac{-5}{\sec \theta + \tan \theta} \right]</math><br/> <math>= 5 \tan \theta.</math> </p> <p> <math>\therefore P</math> is the midpoint of <math>LM</math>.<br/> <math>\therefore LP = PM</math> </p> <p>           (b) (i) <math>I_n = \int_1^2 (\ln x)^n dx</math><br/> <math>= \left[ x (\ln x)^n \right]_1^2 - \int_1^2 \frac{x \cdot n (\ln x)^{n-1}}{x} dx</math><br/> <math>= 2(\ln 2)^n - n I_{n-1}</math> </p> <p>           (ii) <math>I_3 = 2(\ln 2)^3 - 3 I_2</math><br/> <math>= 2(\ln 2)^3 - 3[2(\ln 2)^2 - 2 I_1]</math> </p> <p>           Now <math>I_1 = \int_1^2 \ln x dx</math><br/> <math>= \left[ x \ln x - x \right]_1^2</math><br/> <math>= 2 \ln 2 - 2 - 0 + 1</math><br/> <math>= 2 \ln 2 - 1</math> </p> <p> <math>\therefore I_3 = 2(\ln 2)^3 - 6(\ln 2)^2 + 12 \ln 2 - 6.</math> </p> | <p>           Correctly finding <math>x</math> and <math>y</math> coordinates of midpoint<br/>           Or using distance formula &amp; conclusion → 2 Marks         </p> <p>→ 2 Marks</p> <p>→ 1 Mark</p> <p>→ 1 Mark</p> <p>→ 1 Mark</p> |
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**Question 3**

$$(a) \int \frac{x+1}{x^2+1} dx = \int \left( \frac{x}{x^2+1} + \frac{1}{x^2+1} \right) dx$$

$$= \frac{1}{2} \ln(x^2+1) + \tan^{-1} x + c$$

(b) See next page

(c) (i) Let  $f(x) = x \cos x$  then  $f(-x) = -x \cos(-x)$   
 $= -f(x) \therefore$  it's an odd function.

$$\therefore \int_{-\pi}^{\pi} x \cos x dx = 0$$

$$(ii) \int_0^{\frac{\pi}{2}} \cos^5 \theta d\theta = \int_0^{\frac{\pi}{2}} \cos \theta (\cos^4 \theta) d\theta$$

$$= \int_0^{\frac{\pi}{2}} \cos \theta (1 - \sin^2 \theta)^2 d\theta$$

$$= \int_0^{\frac{\pi}{2}} \cos \theta - 2 \cos \theta \sin^2 \theta + \cos \theta \sin^4 \theta d\theta$$

$$= \left[ \sin \theta - \frac{2}{3} \sin^3 \theta + \frac{1}{5} \sin^5 \theta \right]_0^{\frac{\pi}{2}}$$

$$= 1 - \frac{2}{3} + \frac{1}{5} = \frac{8}{15}$$

$$(d) (i) PF = \sqrt{(x-4)^2 + y^2}$$

$$PM = |x-1| \therefore PF^2 = 4 PM^2$$

$$\therefore x^2 - 8x + 16 + y^2 = 4x^2 - 8x + 4$$

$$\therefore 3x^2 - y^2 = 12 \Rightarrow \frac{x^2}{4} - \frac{y^2}{12} = 1$$

(ii) asymptotes  $y = \pm \sqrt{3}x$

$\therefore$  line  $y = \sqrt{3}x$  makes an angle of  $\frac{\pi}{3}$  rads with positive  $x$ -axis and line  $y = -\sqrt{3}x$  makes an angle of  $\frac{2\pi}{3}$  rads.  $\therefore$  acute angle is  $\frac{\pi}{3}$  rads.

$\rightarrow$  1 Mark

$\rightarrow$  1 Mark

(i)  $\rightarrow$  1 Mark

(ii)  $\rightarrow$  2 Mark

Showing it's an odd function  $\rightarrow$  1 Mark

Answer = 0  $\rightarrow$  1 Mark

$\rightarrow$  1 Mark

Correct integration  $\rightarrow$  1 Mark

Correct answer  $\rightarrow$  1 Mark

$\rightarrow$  1 Mark **each** for  $PF$  and  $PM$  (by definition)

$\rightarrow$   $\frac{1}{2}$  for squaring both sides & expanding

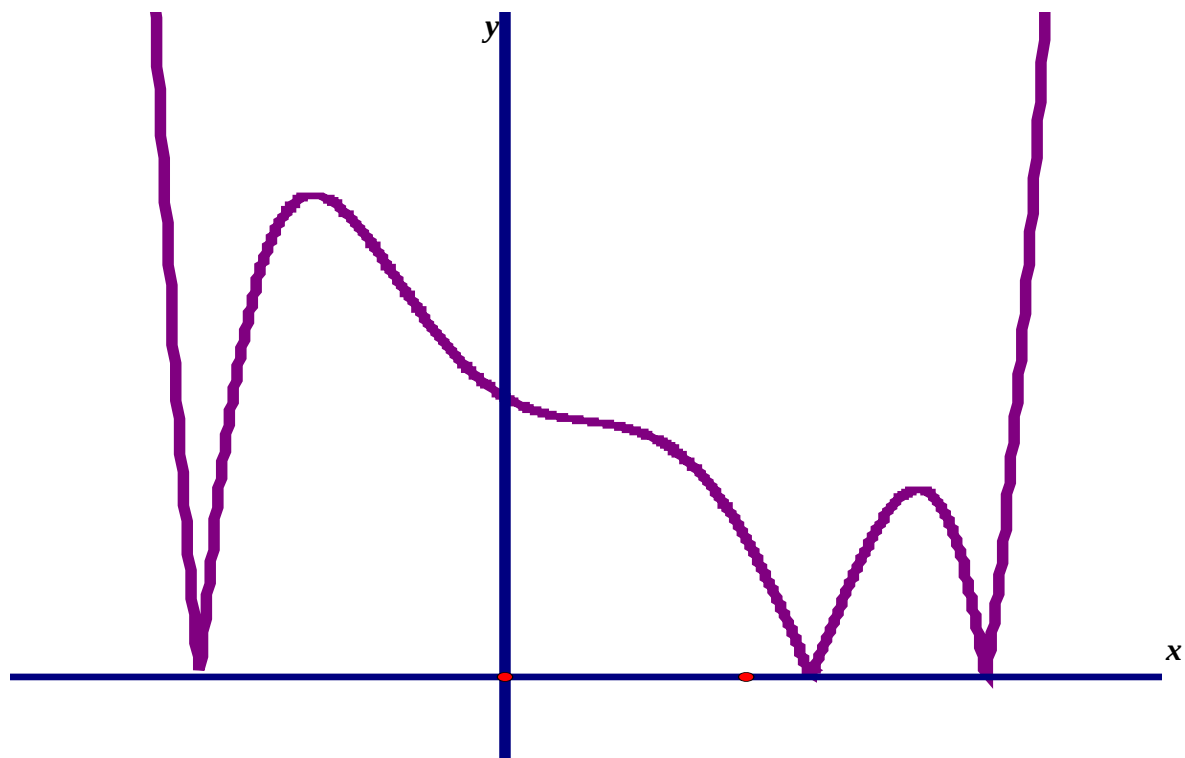
$\rightarrow$   $\frac{1}{2}$  for correct locus.

Correct equations of asymptotes  $\rightarrow$  1 Mark

Correct answer with some explanation  $\rightarrow$  1 Mark

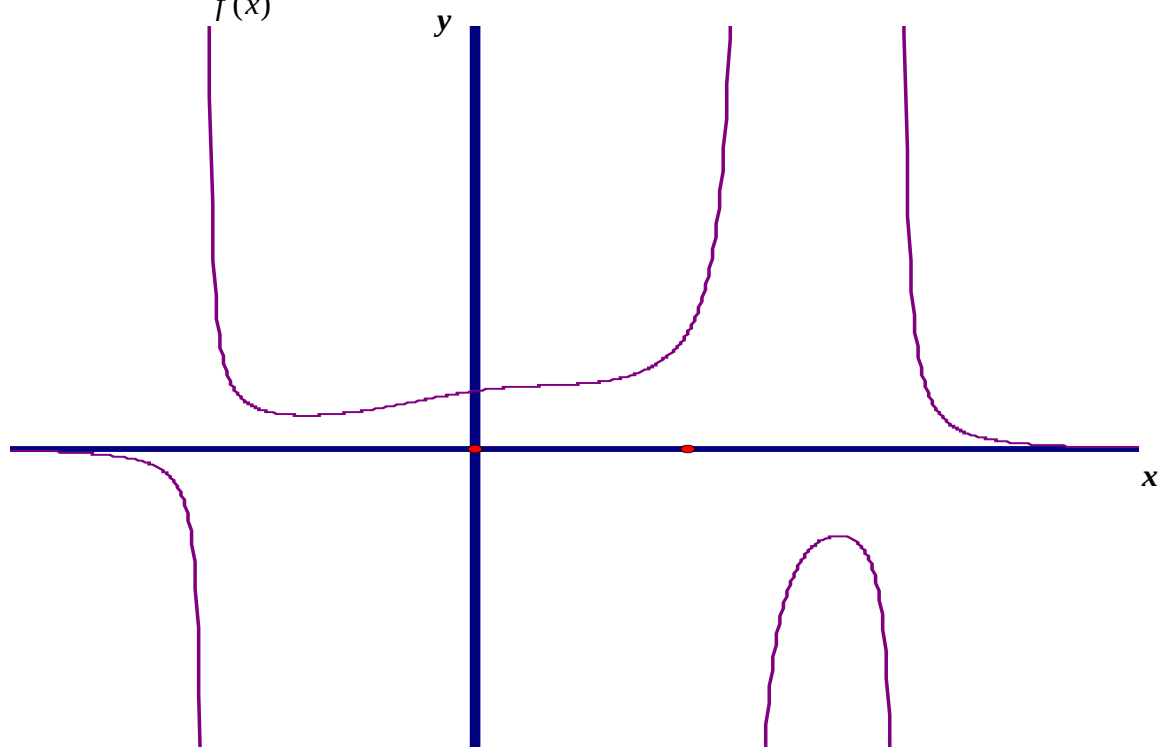
**Q3 (b)** (i)  $y = |f(x)|$

1



(ii)  $y = \frac{1}{f(x)}$

2



**Question 4**

(a)  $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \sin x}$

[when  $x = 0, t = 0$  and  $x = \frac{\pi}{2}$  then  $t = 1$ ]

$$= \int_0^1 \frac{\frac{2}{1+t^2}}{2 + \frac{2t}{1+t^2}} dt$$

$$= \int_0^1 \frac{2 dt}{2 + 2t^2 + 2t} = \int_0^1 \frac{dt}{t^2 + t + 1}$$

$$= \int_0^1 \frac{dt}{\left(t + \frac{1}{2}\right)^2 + \frac{3}{4}}$$

$$= \frac{2}{\sqrt{3}} \left[ \tan^{-1} \left( \frac{t + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) \right]_0^1 = \frac{2}{\sqrt{3}} \left[ \tan^{-1} \left( \frac{2t+1}{\sqrt{3}} \right) \right]_0^1$$

$$= \frac{2}{\sqrt{3}} \left[ \tan^{-1} \sqrt{3} - \tan^{-1} \frac{1}{\sqrt{3}} \right] = \frac{\pi}{3\sqrt{3}}$$

(b) (i)  $\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_a^{2a} f(x) dx$

Now  $\int_a^{2a} f(x) dx = \int_a^0 f(2a-u) \cdot -du$

$$= \int_0^a f(2a-u) du = \int_0^a f(2a-x) dx$$

Since the definite integral is independent of the variable.

Hence  $\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a-x) dx$

$$= \int_0^a [f(x) + f(2a-x)] dx$$

Change of variables & correct  $t$  substitution  
→ 2 Marks

→ 1 Mark

Correct integration & substitution → 1 Mark

→ 1 Mark

Let  $u = 2a - x$  then  $dx = -du$  &  $x = a \rightarrow u = a$  &  $x = 2a \rightarrow u = 0$ . → 1 Mark

→ 1 Mark

$$\begin{aligned}
 \text{(ii)} \quad & \int_0^{\pi} \frac{x \, dx}{2 + \sin x} \\
 &= \int_0^{\frac{\pi}{2}} \left[ \frac{x}{2 + \sin x} + \frac{\pi - x}{2 + \sin(\pi - x)} \right] dx \\
 &= \int_0^{\frac{\pi}{2}} \left[ \frac{\pi}{2 + \sin x} \right] dx \text{ since } \sin x = \sin(\pi - x) \\
 &= \pi \cdot \frac{\pi}{3\sqrt{3}} \text{ from part (a)} \\
 &= \frac{\pi^2}{3\sqrt{3}}
 \end{aligned}$$

Correct use of part (i) → 1 Mark

→ 1 Mark

→ 1 Mark

$$\begin{aligned}
 \text{(c) (i)} \quad & \int_0^{\frac{\pi}{4}} \sec \theta \, d\theta = \left[ \ln(\sec \theta + \tan \theta) \right]_0^{\frac{\pi}{4}} \\
 &= \ln \left( \sec \frac{\pi}{4} + \tan \frac{\pi}{4} \right) - \ln(\sec 0 + \tan 0) \\
 &= \ln(\sqrt{2} + 1)
 \end{aligned}$$

Correct integration → 1 Mark

Correct answer → 1 Mark

$$\begin{aligned}
 \text{(ii)} \quad & \int_0^{\frac{\pi}{4}} \sec^3 \theta \, d\theta = \int \sec \theta \cdot \frac{d}{d\theta} \tan \theta \, d\theta \\
 &= \left[ \sec \theta \tan \theta \right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \tan \theta \sec \theta \tan \theta \, d\theta \\
 &= \sqrt{2} - \int_0^{\frac{\pi}{4}} \sec \theta \tan^2 \theta \, d\theta \\
 &= \sqrt{2} - \int_0^{\frac{\pi}{4}} (\sec^3 \theta - \sec \theta) \, d\theta \\
 &\therefore 2 \int_0^{\frac{\pi}{4}} \sec^3 \theta \, d\theta = \sqrt{2} + \int_0^{\frac{\pi}{4}} \sec \theta \, d\theta \\
 &\qquad\qquad\qquad = \sqrt{2} + \ln(\sqrt{2} + 1) \\
 &\therefore \int_0^{\frac{\pi}{4}} \sec^3 \theta \, d\theta = \frac{1}{2} \left[ \sqrt{2} + \ln(\sqrt{2} + 1) \right]
 \end{aligned}$$

→ 1 Mark

→ 1 Mark

→ 1 Mark using part (i)